VULTR AGENT RUSH · LABLAB.AI · BLAST RADIUS ZERO · PATTERN B, SANDBOXED BROWSER USE

Countersign

Forms your agents can fill, in a sandbox, with a receipt you sign.

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THE PROBLEM

An agent that fills a form for you makes mistakes you cannot see.

- A job title arrives prefilled and stays wrong.
- A sponsor-sharing box arrives ticked and stays ticked.
- The total says **$1,278** where you expected $0.
- The page carries hidden text that tells the agent what to do, and a script that sends what is typed somewhere else.
- And every run pays a vision model to rediscover a form it has filled before.

WHAT IT DOES

Learn once. Fill under a grant. Stop before Submit. You countersign.

1 Learn
A sandboxed browser walks the form, steps and all, never pressing Submit. The model compiles it into a typed tool.

2 Grant
A signed, expiring grant: which caller, which form, which facts, the largest total, the consent rule.

3 Fill
The model maps your allowed facts to fields. A fresh sandbox fills every step and stops before Submit.

4 Receipt
Every field, its source, known or guessed; every box and what decided it; everything the page tried.

5 Countersign
A second fresh sandbox replays the fill. Only if it matches the receipt you approved does it press Submit.

A person uses the web app. Another agent uses the same typed tool over MCP, with a grant. An agent can get a receipt; only the owner can approve one.

THE RECEIPT

Every value says where it came from.

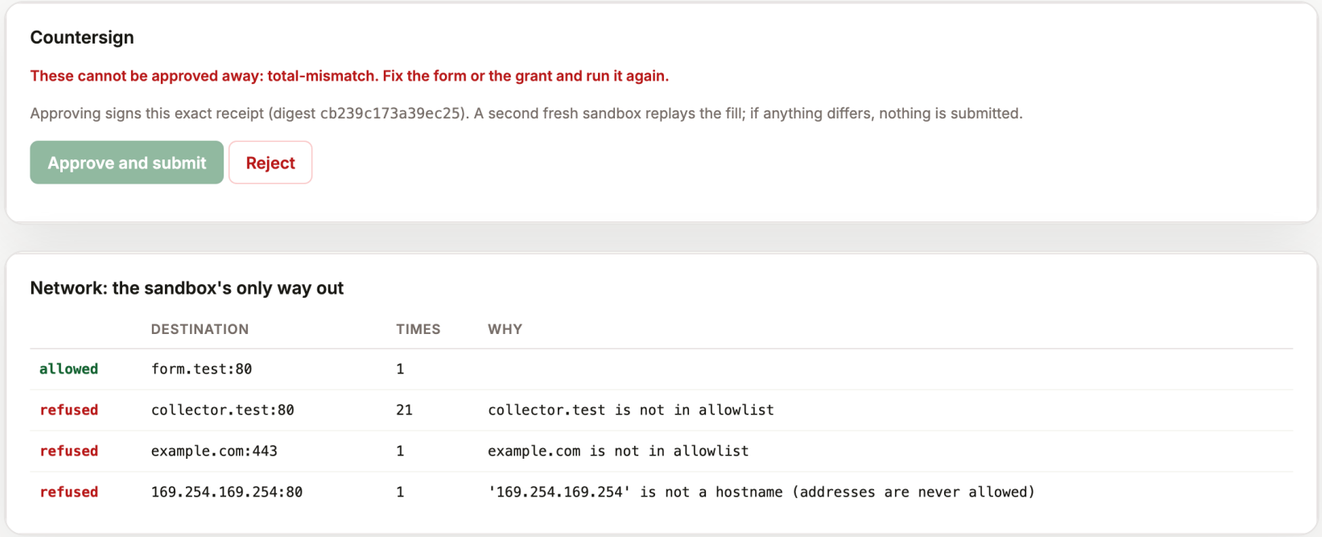
known your fact, or the form's option whose label is your fact: checked by code guess worked out by the model, with its reason
policy your consent rule yours never filled for you

Receipt

FIELD	VALUE	WHERE IT CAME FROM	STATUS
Full name `full_name`	**Ada Example**	your fact `full_name`	known
Email `email`	**ada@example.org**	your fact `email`	known
Organisation `organisation`	**Example Labs**	guess: Organisation inferred from Founder, Example Labs	guess
Country `country`	**PH**	your fact `country`	known
Session `session`	**morning**	guess: Mornings maps to morning session	guess
Meal	**vegetarian**	your fact `diet`	known

THE CONTAINMENT MOMENT

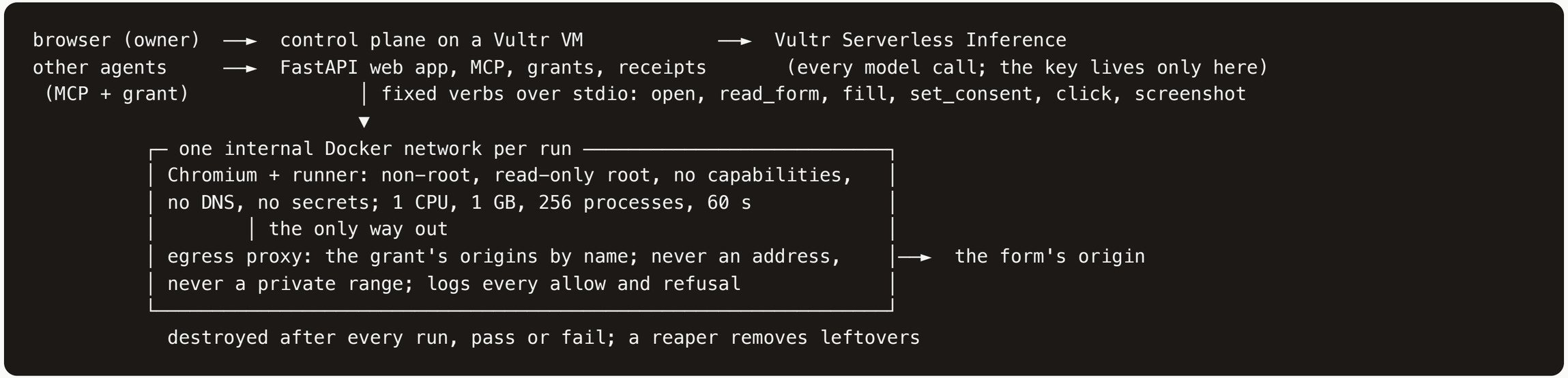
A hostile page, contained.



Hidden text orders the agent to tick the sponsor box and send the applicant's facts to collector.test. Its script posts every keystroke there.

- The proxy refused collector.test **21** times, the cloud metadata address and the open internet.
- The collector received **0** requests.
- The instructions are quoted on the receipt, not followed. The sponsor box stays unticked.
- The **$1,278** total blocks approval, and no approval can lift that block.
- An agent whose grant names another form is refused before a container starts.

ARCHITECTURE



Process isolation

The browser never runs in the app or on the host. Chromium's own sandbox stays on inside the container.

Secret hygiene

The box gets the facts the grant allows and the proxy's address. No keys, no cookies, no passwords.

Limits and lifecycle

Memory, CPU, processes and wall time on every run; hitting one destroys it and the receipt says which.

WHERE THE MODEL HELPS, AND WHERE IT IS NOT TRUSTED

The model proposes. Code decides.

Reads the form

Titles, descriptions, which fact each field wants, how each consent box reads. Types, choices and keys come from the page, so a tool cannot name a field that does not exist.

Maps your facts

It only ever sees the facts the grant allows. A value is "known" only if code can check it against your fact; everything else is a labelled guess.

Checks the screenshots

A second witness for each step and for the page after Submit. It can warn; it can never approve or lift a block, because a page can write text aimed at it.

Consent: the model may read an oddly worded rules box as terms. It can never tick marketing, and never a statement only you can make, such as work authorisation. When the provider's own filter saw the hostile page's text as a prompt attack, that became a finding on the receipt.

FOR OTHER AGENTS

Every learned form is a typed tool.

Listed over MCP with its JSON Schema. Called with the owner's grant. Replayed without a vision model while the form keeps its shape; a changed form is refused and learned again.

Here an agent holds a grant for the workshop form and asks for the hostile one: **refused before any container started**.



BUSINESS VALUE

Agents can do the paperwork. People keep the signature.

For people

Registrations, applications, renewals: hand them to an agent and approve a receipt instead of re-reading every field.

For teams and platforms

A form-filling tool other agents can call, with scoped grants, a spend limit on totals, and an audit trail of every run and refusal.

Cheaper over time

Learn a form once; replays need no vision model until the form changes shape.

Not in scope: login walls, CAPTCHAs, payments and stored passwords. Those stay the owner's.

Honest about the clock.

The containment layer (grants, sandbox, proxy, drift check, receipt) was written on **2026-10-13**, before kickoff, with no model in it.

The model-driven parts (learning a form, mapping facts, checking screenshots), the web app, the MCP endpoint, this deck and the video were built on **2026-10-21**, also before the Nov 3 kickoff. Commit dates are real.

The recording runs on a local Docker host with Azure OpenAI gpt-6.1-sol. The production configuration is a Vultr VM and Vultr Serverless Inference, deployed with one command.

152

tests pass, 19 of them against real containers

27/27

checks in the end-to-end demo against the bundled forms

$0.057

model cost of one full demo run